Alagille Syndrome, Part One: The Genetics. JAG1, NOTCH2, Variable Expressivity, Mosaicism, and Why the Parent Donor Is a Hazard.

Imagine a baby boy, nine weeks old, referred because he is still yellow. His conjugated bilirubin is five point eight milligrams per decilitre, and his gamma-glutamyl transferase, the liver enzyme that rises when bile ducts are injured, is six hundred and ten units per litre. His stools have been pale on some days and normal on others. On examination there is a soft systolic murmur that can be heard in both armpits, which is the typical sound of narrowing in the branches of the pulmonary artery. A cholestasis gene panel comes back with one abnormal result: a heterozygous frameshift variant in the gene called JAG1. His father is well, but he mentions that he had a heart murmur as a child, and the family jokes that everyone on his side has a broad forehead.

The parents ask five questions, and they are exactly the right ones. Does this one result settle the diagnosis, or does the baby still need a liver biopsy or an operation to look at his bile ducts? Did the baby get this from one of us, or did it happen for the first time in him? If both of us test negative, can it happen again in our next child? Can the gene tell us whether he will need a liver transplant? And if he does need one, can his father give part of his liver?

The main idea of this chapter is that Alagille syndrome is a disease of dosage. It is caused by having only about half the normal amount of a signalling protein called Jagged1, or, less often, by a faulty version of its receptor, NOTCH2. Genetic testing now makes the diagnosis in roughly ninety-seven out of every hundred patients. But the genetic result predicts almost nothing about how severe the disease will be, and a parent who looks perfectly normal may carry exactly the same variant as a child who needs a transplant. Every practical decision in this chapter, from counselling the family to choosing a living liver donor, follows from those two facts.

A few terms first. A variant is a change in the DNA sequence; a pathogenic variant is one that causes disease. Heterozygous means that one of the two copies of the gene is changed and the other is normal. Haploinsufficiency means that one working copy is not enough, so that losing half the normal amount of protein is enough to cause disease. A de novo variant is one that has arisen for the first time in the patient and is not present in either parent. Mosaicism means that a person carries a variant in only some of their cells, because it arose after conception; germline mosaicism means that the variant is present in the egg or sperm cells, sometimes without being detectable in blood. Penetrance is the proportion of carriers who show any sign of the condition, and expressivity is how severe those signs are. Bile duct paucity, also called ductopenia, means too few bile ducts in the portal tracts of the liver on biopsy.

The evidence in this chapter comes from several sources. The gene was discovered in 1997 by two groups, Li and colleagues and Oda and colleagues, both publishing in Nature Genetics. The second gene, NOTCH2, was reported by McDaniell and colleagues in the American Journal of Human Genetics in 2006. The largest single-centre genetic description is the mutation update by Gilbert

and colleagues from the Children's Hospital of Philadelphia, published in Human Mutation in 2019, and the largest international comparison of the two genes comes from the Global Alagille Alliance, known as GALA, reported by Vandriel and colleagues in Liver International in 2025. Family studies come from Kamath and colleagues in the Journal of Medical Genetics in 2003, and from Giannakudis and colleagues in the European Journal of Human Genetics in 2001. The living donor data come from the Kyoto series by Kasahara and colleagues in Transplantation in 2003, and a 2023 review by Kasahara, Hong and Dhawan in the Journal of Hepatology. Guidance comes from the 2017 joint guideline on cholestatic jaundice in infants from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, from the 2024 Clinical Practice Guidelines of the European Association for the Study of the Liver on genetic cholestatic liver diseases, and from the GeneReviews chapter on Alagille syndrome, last revised in 2024. For brevity, these will be called the 2017 joint guideline, the European guideline, and GeneReviews.

Part one. How Jagged1 builds a bile duct

To understand why a single gene causes problems in the liver, the heart, the spine, the eyes and the kidneys, it helps to follow one signal step by step.

The first step is the ligand. Jagged1 is a protein that sits on the surface of cells. In the developing liver it is made by the mesenchyme around the portal vein, including the smooth muscle cells that are forming the vessel wall. Because Jagged1 is anchored in the cell membrane, it can only signal to cells that it is physically touching. This kind of contact signalling is called juxtacrine signalling.

The second step is the receptor. The cells touching the portal mesenchyme are hepatoblasts, the immature cells that can become either liver cells or bile duct cells. These hepatoblasts carry a receptor called NOTCH2. When Jagged1 binds NOTCH2, the signalling cell pulls on the receptor, and this mechanical pull exposes a cutting site on NOTCH2. Two enzymes then cut the receptor in sequence: first a protease called ADAM10, and then the enzyme complex called gamma-secretase.

The third step is the signal itself. The second cut releases the inside part of the receptor, called the Notch intracellular domain. This fragment travels into the nucleus, binds a DNA-binding partner called RBPJ, and switches on target genes, including HES1, HEY1 and SOX9.

The fourth step is a change of fate. Hepatoblasts that receive this signal become biliary cells. They form a single layer around the portal vein, called the ductal plate, and switch on biliary identity genes such as SOX9 and hepatocyte nuclear factor one beta.

The fifth step is building the tube. The ductal plate remodels into a proper duct with a central channel, or lumen. If the signal is too weak, ducts either never form properly or form and are then lost. That is what the pathologist sees as bile duct paucity.

Mouse experiments support each step. When Hofmann and colleagues, writing in Development in 2010, deleted Jagged1 only in the portal vein mesenchyme, the mice developed bile duct paucity, showing that the signal comes from the mesenchyme. When McCright and colleagues, also in

Development in 2002, made mice that had only one working copy of both Jagged1 and NOTCH2, the animals developed jaundice, too few bile ducts, and defects of the heart, eye and kidney, which together look very much like the human syndrome. The same pairing of Jagged1 and Notch receptors also patterns the pulmonary arteries, the vertebrae, the front of the eye and the kidney. That explains why one gene gives a syndrome that affects so many organs.

Part two. Why half a dose is enough

There are four lines of evidence that JAG1 disease is caused by haploinsufficiency, that is, by having only half the normal amount of working protein.

First, the type of variant. In the Philadelphia mutation update, eighty-three percent of all reported disease-causing JAG1 variants either cut the protein short or remove it altogether. A variant that introduces an early stop signal usually causes the messenger RNA to be destroyed by a cellular quality-control system called nonsense-mediated decay, so that the faulty copy makes no protein at all.

Second, whole-gene deletions. Some patients have lost the entire JAG1 gene on one chromosome. They have the same features as patients with small changes inside the gene. If losing the whole copy gives the same disease as a small change, the small change must be acting simply by removing that copy's function.

Third, the mouse genetics already described. One faulty copy of both Jagged1 and NOTCH2 together produces the syndrome, whereas mice with only one faulty Jagged1 copy on mixed genetic backgrounds are mildly affected, mainly in the eye. The dose of the whole pathway matters.

Fourth, the genetic background. Thakurdas and colleagues, writing in Hepatology in 2016, showed that mice with one faulty Jagged1 copy on a particular inbred background develop a true cholangiopathy around birth, and that removing one copy of a second gene, called POGLUT1 or Rumi, which adds sugar groups to Notch receptors, rescues the bile ducts. In other words, other genes can move the threshold.

Put simply, most patients make about half the normal amount of functional Jagged1. The system sits close to a threshold, so small differences in other genes, in the sugar decoration of Notch, or in plain chance during development decide whether a duct forms. That is the biological basis for the variable severity that we will meet again and again in this chapter.

Part three. The genetic architecture

How often does testing find the cause? In the Philadelphia cohort of four hundred and one probands, meaning the first affected person in each family, collected over twenty-seven years, a JAG1 variant was found in ninety-four point three percent, a NOTCH2 variant in two point five percent, and no variant in three point two percent. The international GALA cohort of nine hundred and fifty-two patients from sixty-six centres in twenty-nine countries found almost the same split:

ninety-four point seven percent JAG1, three point six percent NOTCH2, and one point seven percent with neither. The 2017 joint guideline uses the round figures of about ninety-five percent and five percent.

Families are almost always unique. The Philadelphia group found two hundred and ninety-seven different pathogenic JAG1 variants in their own patients, and six hundred and ninety-four across the published literature. There is no common mutation to test for, so the whole gene must always be examined.

The first NOTCH2 families were found when McDaniell and colleagues sequenced the gene in eleven patients who clearly had the syndrome but had no JAG1 variant. Two families carried NOTCH2 variants, five affected people in all, and every one of them had kidney disease, which is usually a minor part of the syndrome.

What about the small group in whom nothing is found? Some carry variants deep inside the non-coding parts of JAG1, some are mosaic, some have structural rearrangements that standard tests do not see, and some probably have a different diagnosis. The practical lesson is that a clinically convincing patient with a normal JAG1 sequence is not truly mutation-negative until copy number and NOTCH2 have both been examined.

Part four. The variant spectrum and the problem of deletions

In the Philadelphia cohort, the JAG1 variants were distributed as follows. Frameshift variants, which shift the reading frame of the gene, made up thirty-seven percent. Nonsense variants, which create a premature stop signal, made up twenty-two percent. Large deletions removing the whole gene made up thirteen percent. Missense variants, which change a single amino acid, also made up thirteen percent. Splice-site variants, which disrupt the joining of the gene's coding pieces, made up twelve percent. In-frame deletions, duplications, inversions and translocations together made up the remaining few percent.

The thirteen percent of whole-gene deletions matters at the bedside. Standard sequencing reads the letters of the gene, but if a whole copy is missing, the remaining copy reads normally and the test looks clean. According to GeneReviews, sequencing detects about eighty-eight percent of the JAG1 variants, and a separate test for deletions and duplications finds the other twelve percent. That separate test can be multiplex ligation-dependent probe amplification, quantitative polymerase chain reaction, or a chromosomal microarray. Many modern panels call these copy-number changes from the sequencing data, but not all do, and some exome pipelines miss them. So when ordering a panel, ask the laboratory directly whether it reports copy-number changes in JAG1. A test that misses deletions misses roughly one patient in eight.

Some deletions are large enough to remove neighbouring genes on the short arm of chromosome twenty, in the band called twenty p twelve. Children with these contiguous gene deletions may have developmental delay, hearing loss or autism in addition to the syndrome. GeneReviews advises that developmental delay or hearing loss in a child with Alagille syndrome should prompt a

chromosomal microarray to size the deletion.

Part five. Missense variants, the hard calls

Truncating variants and deletions are easy to interpret, because they clearly destroy the protein. Missense variants are harder, because a single amino acid change can be harmless or devastating. Across the literature, one hundred and four of the six hundred and ninety-four JAG1 variants, about fifteen percent, are missense.

Three patterns help. First, disease-causing missense variants cluster in the first six exons of the gene, which encode the signal peptide that guides the protein to the cell surface, the DSL domain that binds the receptor, and the first two of the protein's repeating units called EGF-like repeats. Second, EGF-like repeats are held in shape by bonds between pairs of cysteine amino acids. A variant that removes a cysteine from one of these repeats is enriched in patients, whereas a variant that adds a new cysteine is more common in healthy population databases such as gnomAD. Third, when the Philadelphia group applied the standard American College of Medical Genetics and Genomics criteria to thirty-eight missense variants, six could not be confidently classified and remained variants of uncertain significance. Functional tests in cells helped: most cysteine-losing variants left the protein trapped inside the cell, near the nucleus, so that it never reached the surface to signal.

The mouse model called Nodder, described by Andersson and colleagues in Gastroenterology in 2018, carries a single amino acid change, histidine two hundred and sixty-eight to glutamine. Its Jagged1 cannot bind NOTCH1 at all and binds NOTCH2 only weakly. These mice have liver, heart, eye and facial features of the syndrome, and interestingly, their bile ducts partly recover in adulthood.

What should the clinician do with a variant of uncertain significance? The European guideline accepts that a liver biopsy can be useful in exactly this situation, together with a careful search for other features of the syndrome: an echocardiogram, an eye examination looking for posterior embryotoxon, and a spine film looking for butterfly vertebrae. Testing the parents to see whether the variant tracks with features in the family also helps.

Part six. NOTCH2 disease looks different

The GALA study compared nine hundred and two patients with JAG1 variants to thirty-four with NOTCH2 variants, and the picture was strikingly different outside the liver. The characteristic face was present in ninety point one percent of JAG1 patients but only fifty-seven point six percent of NOTCH2 patients. A cardiac anomaly on echocardiogram was found in ninety-two point two percent versus sixty-four point three percent. Posterior embryotoxon, a visible white ring at the edge of the cornea, was found in fifty-two point eight percent versus eighteen point five percent. Butterfly vertebrae, in which a vertebral body is split down the middle, were found in forty-four point five percent versus only three point three percent. All of these differences were highly significant. In contrast, kidney anomalies were similar, at thirty-nine point two percent versus

thirty-four point five percent, and so were the liver features: neonatal cholestasis in eighty-three percent versus ninety-one percent, bile duct paucity on the first biopsy in sixty-five percent versus fifty-seven percent, and any liver involvement in about ninety-nine percent versus one hundred percent. Survival with the native liver, and overall survival, did not differ between the two genes. One more surprise was sex: seventy-nine percent of the NOTCH2 patients were male, compared with fifty-six percent of the JAG1 patients.

There is an important caveat. The GALA patients were recruited from liver clinics, so every one of them was found because of liver disease; NOTCH2 carriers with mainly kidney disease might be missed. But the clinical message is clear. A cholestatic infant with a normal face and a normal spine can still have Alagille syndrome, and the authors warn that relying on the classical clinical definition may miss patients with NOTCH2 disease. Testing should be broad.

Part seven. NOTCH2 variants are harder to read

NOTCH2 variants are also harder to interpret than JAG1 variants. In the Philadelphia review, sixty-eight percent of known NOTCH2 variants were missense, compared with about thirteen percent for JAG1, so the simple rule that a truncating variant equals loss of function does not carry over. Eighty-five percent of NOTCH2 missense variants sit in two hubs: the EGF-like repeats encoded by exons seven and eight, and the ankyrin repeats, which are part of the signalling machinery inside the cell, encoded by exons thirty-one and thirty-two.

When the GALA group re-examined sixty-one previously published NOTCH2 variants using disease-specific rules, nineteen of them, thirty-one percent, were downgraded from pathogenic or likely pathogenic to uncertain significance, and ten more, sixteen percent, were reclassified as likely benign. Only thirty-four percent kept their original label.

The position of a NOTCH2 variant matters enormously. Truncating variants in the very last exon of NOTCH2 escape nonsense-mediated decay, because that quality-control system cannot act on the final exon. They produce a stable, overactive intracellular fragment and cause a completely different disorder, Hajdu-Cheney syndrome, a condition of bone loss and skeletal abnormalities. This was shown by Simpson and colleagues and by Isidor and colleagues, both in Nature Genetics in 2011. So the same gene gives loss of function with one variant and gain of function with another. The practical rule is that a NOTCH2 report that says likely pathogenic and dates from before 2025 deserves reinterpretation against the GALA criteria.

Part eight. The same variant, a different disease in every relative

The most important clinical feature of Alagille genetics is variable expressivity. Kamath and colleagues, in the Journal of Medical Genetics in 2003, studied thirty-four probands and then systematically examined fifty-three of their relatives who carried the same JAG1 variant. Only eleven of these fifty-three relatives, twenty-one percent, had features that would have led to a diagnosis on their own. Seventeen, thirty-two percent, had mild features that were found only because the doctors went looking for them. Twenty-five, forty-seven percent, did not meet clinical

criteria at all, and two of these had no features whatsoever. That gives a penetrance of about ninety-six percent if any feature counts, but a much lower penetrance for clinically important disease. The most consistent feature in carriers was the face, which was almost universal; heart and liver disease were less common in relatives than in probands.

A study from the Exeter genetics laboratory by Guegan and colleagues, published in Clinical Genetics in 2012, looked at the same question from the other direction. Among one hundred and thirty-five probands referred for testing, many with only a partial picture, a JAG1 variant was found in fifty-nine, forty-four percent. The detection rate rose from about twenty percent in patients with only one affected organ system to eighty-six percent in those with all five, and about a third of patients with only one or two features still had a JAG1 variant.

The lesson is that the classic diagnostic criteria were written for probands, the sickest member of each family. Among relatives, almost half of carriers would never be diagnosed clinically. And relatives are exactly the people who come forward as living liver donors.

Part nine. Why genotype does not predict the liver

Neither the type nor the position of a JAG1 or NOTCH2 variant predicts liver disease, heart disease or survival. Whole-gene deletions behave like small point variants, and there is no mutational hotspot that marks a severe course. What, then, explains why one sibling needs a transplant and another has normal liver tests?

Several explanations have been proposed. One is modifier genes. Tsai and colleagues, writing in Cellular and Molecular Gastroenterology and Hepatology in 2016, performed a genome-wide association study in one hundred and sixty-one JAG1 carriers of European ancestry, comparing ninety-seven with mild liver disease to sixty-four with severe liver disease. The strongest signal lay near a gene called THBS2, thrombospondin two, with a p value of one point eight times ten to the minus six. That is suggestive, but it did not reach the stricter threshold for genome-wide significance. In laboratory work, thrombospondin two bound Notch receptors and blocked Jagged1 from binding NOTCH2, which makes it a plausible modifier, but it remains a candidate rather than a proven one. A second explanation is the enzymes that decorate Notch with sugars, including POGLUT1 and the Fringe enzymes, which, as we saw in mice, can shift the threshold. A third is chance. Kamath and colleagues reported monozygotic twins with the same JAG1 deletion whose heart disease was discordant, in the American Journal of Medical Genetics in 2002; identical twins share every gene, so the difference must come from stochastic events in development. A fourth is time: bile duct paucity may be absent in a young infant and appear later, while some children's liver tests improve with age, much as the Nodder mice partly regenerate their ducts.

The practical point is never to use a sibling's or a parent's course to forecast a child's liver. Prognosis comes from the child's own trajectory, especially bilirubin and other markers in the first year of life, which will be the subject of a later lesson.

Part ten. Inheritance, mosaicism and the next pregnancy

Alagille syndrome is inherited in an autosomal dominant way. According to GeneReviews, about sixty percent of probands have a de novo variant and about forty percent have inherited it from a parent. Because of variable expressivity, a parent who considers himself unaffected is often a mildly affected carrier, as the father in our opening case may be.

Mosaicism adds a further layer. Giannakudis and colleagues screened families with JAG1 variants and found that in five of sixty-one, eight point two percent, a parent was mosaic for the variant, carrying it in only some cells. They regarded this as a minimum estimate. Mosaic parents are often very mildly affected, which makes the diagnosis and the risk estimate even harder.

What does this mean for the next pregnancy? Each child of a carrier has a fifty percent chance of inheriting the variant, whatever the parent's own severity, and the severity in that child cannot be predicted. If both parents test negative in blood, the risk to a future child is lower but is not zero. A blood test can miss a low level of mosaicism, and it cannot detect a variant that is confined to the egg or sperm cells. GeneReviews therefore describes the risk to siblings in that situation as greater than that of the general population. Brothers and sisters of a child with an apparently de novo variant should still be offered testing.

Once the family variant is known, prenatal testing and preimplantation genetic testing are possible. But families need to know two things. A positive result says nothing about how severe the disease will be. And a normal fetal echocardiogram does not exclude the syndrome.

Part eleven. A practical testing pathway

Putting this together gives a five-step approach.

The first step is to suspect the diagnosis. A cholestatic infant with a high gamma-glutamyl transferase and any of a heart murmur, an unusual face, posterior embryotoxon, a butterfly vertebra or a kidney anomaly should have an echocardiogram, an eye examination and a spine film early.

The second step is a gene panel. The 2017 joint guideline recommends a low threshold for gene panels or exome sequencing in cholestatic infants and states that the diagnosis of Alagille syndrome is confirmed by sequencing JAG1 and NOTCH2. The European guideline adds that genetic counselling should come before testing and that next-generation sequencing is the recommended method. The panel should include both genes and should call copy-number changes. Ideally, the result should be available before any operative cholangiogram.

That last point deserves emphasis. Infants with Alagille syndrome can have a thin, hypoplastic extrahepatic bile duct system that can be mistaken for biliary atresia, and a Kasai operation in an Alagille infant does not help and can worsen the outcome. The 2017 joint guideline advises looking for features of the syndrome before an intraoperative cholangiogram.

The third step applies when the panel is negative and suspicion remains strong. Then the next tests are a targeted deletion test or microarray, followed by exome or genome sequencing, with a request to look at non-coding parts of JAG1 and to reanalyse the data over time.

The fourth step applies when the result is a variant of uncertain significance. Here the clinician tests the parents to see whether the variant tracks with features, searches carefully for other features, and looks for functional data. This is the situation in which the European guideline considers liver biopsy useful. It also stresses that paucity is relatively insensitive in young infants and does not recommend routine biopsy to determine prognosis.

The fifth step applies when the result is positive. Both parents should be tested and cascade testing offered to the wider family. The exact variant should be recorded clearly in the notes, because it will be needed if a related living donor is ever considered.

Part twelve. Why the parent is the riskiest donor

Living donor liver transplantation usually uses a parent, and in Alagille syndrome that creates a specific danger. The parent is both the most likely donor and the most likely carrier.

The Kyoto experience, reported by Kasahara and colleagues in 2003, shows what is at stake. Between 1990 and 2002, twenty children with Alagille syndrome, with a median age of five years, received grafts from living relatives. Five potential parental donors were excluded because their own liver biopsy showed bile duct paucity, and one more was excluded because of an abnormality of the hepatic blood vessels. Even so, three recipients died: one from surgical complications, one from heart failure caused by progressive pulmonary artery stenosis, and one because the graft itself had unsuspected bile duct paucity. Five-year patient survival was eighty point four percent. In the children who did well, liver function improved in every case, and ninety percent showed catch-up growth.

There are three reasons why a carrier parent is a hazardous donor. First, the graft may already be ductopenic, so the child receives a liver with the same disease. Second, the donor may have vascular anomalies that belong to the syndrome, which raise the donor's own surgical risk. Third, because severity cannot be predicted, a donor who looks well today may have a liver that does not tolerate a major hepatectomy. The 2023 Journal of Hepatology review of living donors for hereditary liver disease lists Alagille syndrome among the few heterozygous conditions in which parental donation has caused harm to the recipient, and recommends genetic testing as part of parental donor selection.

Part thirteen. How to work up a related donor

GeneReviews and the transplant literature suggest the following approach. If the child's variant is known, the potential donor should have a targeted test for that variant. A donor who carries the variant should be excluded, because roughly half of carriers look normal and the graft may be affected. A donor who tests negative in blood should still be examined, with attention to the face, an echocardiogram, an eye examination, a spine film and liver tests, because low-level or tissue-specific mosaicism can be missed by a blood test. If the child's variant is not known, the donor needs a full clinical screen, and a liver biopsy of the donor should be considered; the Kyoto team excluded five donors on biopsy findings alone. The donor's blood vessels should be imaged

with computed tomography or magnetic resonance angiography. And when there is any doubt, an unrelated or deceased donor removes the inherited-graft question entirely.

A simple rule summarises this. No related donor should proceed without either a negative targeted test together with a normal examination, or, when the variant is unknown, a normal examination and a normal donor biopsy.

Part fourteen. What the guidelines say

How do the major documents compare on the genetics of Alagille syndrome? On the first-line test, they agree. The 2017 joint guideline, which serves as both the North American and the European paediatric position, recommends a low threshold for panels or exome sequencing and confirmation by JAG1 and NOTCH2 sequencing. The 2024 European guideline recommends next-generation sequencing preceded by genetic counselling. This is aligned.

On liver biopsy, the emphasis differs. The 2017 guideline still places biopsy within the general work-up of the cholestatic infant and advises looking for Alagille features before a cholangiogram. The 2024 European guideline goes further: it does not recommend routine biopsy to determine prognosis or management, it notes that paucity is insensitive in young infants, and it limits biopsy to situations such as a variant of uncertain significance. This is a real difference in emphasis.

On several questions there is simply no guidance. None of the documents says explicitly that copy-number testing is required, none offers rules for interpreting NOTCH2 variants, although the GALA criteria now exist in the literature, and none addresses parental mosaicism or recurrence risk beyond a general statement about counselling. These are gaps.

The most material gap concerns the related living donor. No society guideline grades a recommendation on screening a related donor for Alagille syndrome. The European guideline encourages referral to a transplant centre for children with significant liver disease but says nothing about donor selection. Yet the Kyoto experience includes a recipient death from an unsuspected ductopenic parental graft. Current practice therefore rests on GeneReviews, the 2023 review, and the protocols of individual transplant centres.

Part fifteen. Common traps

Several errors recur in practice. One is calling a patient JAG1-negative after sequencing alone, when twelve to thirteen percent of JAG1 variants are whole-gene deletions. Another is stopping at JAG1, when NOTCH2 explains about three percent of patients, many of whom lack the classic face, eye and spine findings. A third is requiring the full classic criteria in a relative, when half of carrier relatives would never meet them. A fourth is using a parent's mild course to reassure the family about the child, or a sibling's transplant to frighten them, when there is no genotype-phenotype rule. A fifth is telling parents who both test negative that recurrence is impossible, when parental mosaicism was found in about one family in twelve and germline mosaicism is invisible in blood. A sixth is accepting an old NOTCH2 report without

reinterpretation, when a third of published NOTCH2 variants were downgraded in 2025. A seventh is waiting for paucity on biopsy in a young infant, when it may not be there yet and genetics gives the answer sooner. An eighth is accepting a related living donor on the basis of normal liver tests alone. A ninth is ordering a panel without asking whether it reports copy-number changes, and failing to record the exact variant for a future donor work-up. And a tenth is missing a large twenty p twelve deletion in a child with developmental delay or hearing loss, when a microarray would show it.

Part sixteen. What to do differently in clinic

Let us return to the nine-week-old boy and answer his parents' five questions.

Does the result settle the diagnosis? A truncating JAG1 variant in a cholestatic infant with a murmur is strong molecular confirmation. He still needs an echocardiogram, an eye examination, a spine film and a kidney ultrasound to define the extent of his disease, but he does not need a biopsy to make the diagnosis, and he should not have a cholangiogram or a Kasai operation on the assumption that this is biliary atresia.

Did the variant come from a parent? The father's childhood murmur and the family forehead are clues, not a diagnosis. Both parents should be tested for the specific variant. If the father carries it, he is a mildly affected carrier, and each future child has a fifty percent chance of inheriting it, with severity that cannot be predicted. If both parents test negative, the variant is probably de novo, but the risk to a future child is still above that of the general population because of possible germline mosaicism. Prenatal or preimplantation testing can be offered once the variant is known.

Can the gene predict his liver? No. The type of variant does not predict the course, and neither does the father's mild picture. His own bilirubin, itch, growth and cholesterol over the first year will guide the prognosis.

Can his father be a donor? Only if he tests negative for the family variant and has a normal examination, echocardiogram and vascular imaging. If he carries the variant, he should not donate, and the team should look for an unrelated living donor or a deceased donor graft.

The broader changes for clinic are these. Order a panel that includes both JAG1 and NOTCH2 and that calls copy-number changes, and send it early, before any operation on the bile ducts. Treat a negative sequence as incomplete until deletions and NOTCH2 have been checked. Reinterpret old NOTCH2 reports. Examine relatives rather than accepting that they are well. Never tell a family that recurrence is impossible. Do not predict a child's liver from a relative's. And record the exact variant in the notes, because one day it may decide whether a parent can safely give part of their liver. The gene makes the diagnosis; the child's own course makes the prognosis; and the related donor is tested, not trusted.